

Dual-Precision Deep Neural Network

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ABSTRACT

On-line Precision scalability of the deep neural networks(DNNs) is a critical feature to support accuracy and complexity trade-off during the DNN inference. In this paper, we propose dual-precision DNN that includes two different precision modes in a single model, thereby supporting an on-line precision switch without re-training. The proposed two-phase training process optimizes both low- and high-precision modes.

Keywords

Deep neural network; weight quantization; dual precision; precision scalable.

1. INTRODUCTION

Deep Neural Networks (DNNs) require a large number of parameters and operations to be stored and computed, which act as a critical challenge of accelerating DNN inference on the edge devices with limited resources. One of the solutions for this challenge can be utilizing the quantization method that reduces the numerical precision of the weight parameters and/or activations, which determines the accuracy and required compute/memory resources of the inference. As the accuracy requirement and available resources can vary due to the nature of dynamic environment of the edge device applications, a critical feature for DNN quantization is on-line precision scalability of DNN models, supporting dynamic trade-off between accuracy and complexity of the DNN inference.

One solution for this demand is to store two separate DNN models trained for different target precision (b -bit and $(b+1)$ -bit precision), and apply either of the two depending on the accuracy/complexity requirements (Fig. 1(a)). However, this solution requires additional memory for storing the two DNN models. Another solution can be transforming a DNN model on-demand for the specific target precision as in Fig. 1(b). However, the existing precision scaling methods generally require retraining of the DNN models, which limits an on-line switch of the model precision. Although post quantization techniques can be applied to remove the need for re-training, the accuracy cannot be guaranteed when down-scaling into the low precision.

In this paper, we propose a new method to train dual-precision DNNs that can switch between the two precision modes without the need for re-training. By sharing the common bits (b bits) between the low- and high-precision weights, the dual-precision DNN allows dynamic on-line switch between the two precision modes only with truncation (precision down-scaling) or concatenation (precision up-scaling) of the last 1 bits. To optimize both precision modes in a single model, we propose the training process with two phases - first phase for shared-bit training and second phase for the entire high-precision training. After applying the proposed training

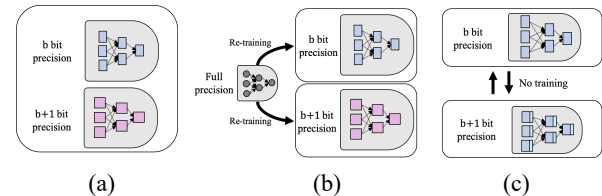


Figure 1. Solutions for on-line precision switch of a DNN model on a device. (a) a device contains two separate models, each trained for the target precision, (b) a device contains a single model, but re-trained on-demand for the target precisions, (c) the proposed dual-precision DNN that contains two different precision modes in a single model, supporting easy switch between different precision modes.

method, the first b bits and the entire $b+1$ bits of the weights are optimized for the low-precision and high-precision modes, respectively, each showing comparable accuracy to the baseline models trained for the target precision using the existing quantization method.

2. RELATED WORK

In recent years, a growing number of studies have proposed techniques for reducing the bit precision of DNN models.

One line of studies proposed DNN training methods to minimize accuracy loss under quantization of weights and/or activations. DNNs such as BinaryConnect [4], BNN [7], and XNOR-Net [16] have been proposed to quantize weights/activations to 1-bit. TWN [8,13] added the level zero, and other studies [5,19] have represented parameters as multiple levels expressed as integers using the rounding function. To exploit the bell-shaped distribution of the weights, LogNet [12] represented quantized parameters in the log domain. Another line of studies [6,21] applied re-training after quantization to compensate accuracy drop. As the above quantization methods are tightly coupled with the training process, changing the model precision requires retraining of the model, otherwise the accuracy drop can be significant [3].

To enable precision down-scaling without training, several studies have proposed post quantization techniques, which mainly focus on finding the clipping threshold. Sung et al. [18] used L2 error to optimize fixed-point quantization, and Migacz et al. [15] found the minimum value using the KL divergence between the original distribution and the quantized distribution. Banner et al. [1] used Gaussian and Laplacian distribution to model the prior distribution, and Zhao et al. [20] divided the channel of filters to reduce the size of the clip value.